

ECHO Unit Queue-Management System

Administrator Web Portal — Screen-by-Screen UI Documentation

Prepared from the submitted mockup image, cross-checked against the Research Proposal (Chapters 1, 3 and 5)

1. Introduction

This document reviews the interface mockups supplied for the ECHO Unit queue-management system and matches each screen to the correct actor (Patient, Staff, or Administrator), a description of its purpose, and its concrete functions. The mockup image contains twelve distinct screens/states. Ten of these are clearly Administrator-side screens, consistent with the sidebar navigation labelled "ECHO Unit Admin" and the Super Admin account shown in the footer of most screens. However, two of the twelve screens are not admin-specific — they are the system's shared entry points used by all three actor types (Patient, Staff, and Administrator). These are flagged explicitly below so the documentation accurately reflects what was actually designed, rather than assuming every screen in the strip belongs to the admin portal.

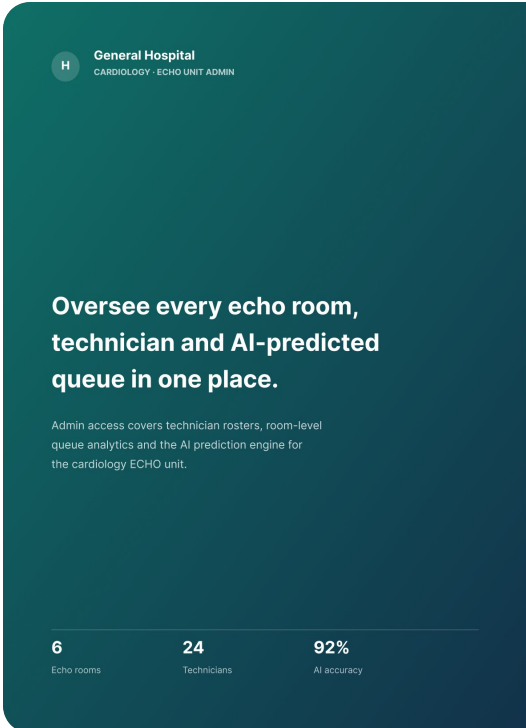
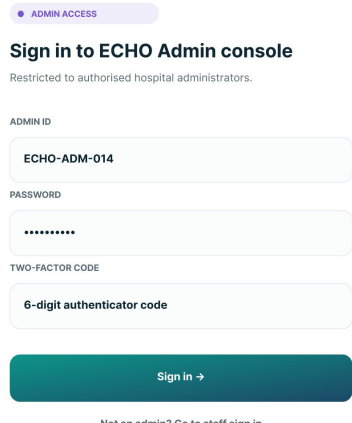
The Research Proposal (Section 1.5, Scope of the Study) states that the Administrator Web Portal consists of eight core screens: a Today's Unit Overview dashboard, a Room/Station Management grid, a Staff Accounts table, an Add-Staff-Member modal, an Audit Log, a Settings screen, a Register Station modal, and an Add Echo Room modal. The mockups largely match this list, but the implementation actually contains a few additional screens/states (an Admin sign-in screen, an Admin "My Profile" account screen with an edit-profile sub-view, a quick "Add Technician" modal, and a detailed staff "Access & Role" modal) that are not named individually in the proposal text. These additions and the minor naming differences are called out under "Notes / Corrections" in the relevant sections.

2. Screen Inventory at a Glance

#	Screen	Actor	Proposal screen it fulfils
1	Admin Sign-In	Administrator	Not individually named — required entry point
2	Today's Unit Overview (Dashboard)	Administrator	Today's Unit Overview dashboard
3	Echo Rooms (Room/Station grid)	Administrator	Room/Station Management grid
4	Staff Accounts	Administrator	Staff Accounts management table
5	Staff Member — Access &	Administrator	Extends Add-Staff-Member modal

#	Screen	Actor	Proposal screen it fulfils
	Role modal		
6	Audit Log	Administrator	Audit Log
7	Settings	Administrator	Settings screen
8	Add Technician (quick-add modal)	Administrator	Variant of Add-Staff-Member modal
9	Add Echo Room modal	Administrator	Add Echo Room modal (also stands in for "Register Station")
10	Add Staff Member modal	Administrator	Add-Staff-Member modal
11	My Profile / Edit Profile	Administrator	Not named in proposal — supporting account screen
12	Unified Sign-In (role selector)	Patient / Staff / Administrator (shared)	Not admin-specific — shared login gateway

Screen 1: Admin Sign-In

Primary Actor	Administrator
 The left panel of the sign-in screen features a dark teal background. At the top left is a circular logo with the letter 'H' and the text 'General Hospital' and 'CARDIOLOGY - ECHO UNIT ADMIN'. The main heading reads 'Oversee every echo room, technician and AI-predicted queue in one place.' Below this is a descriptive paragraph: 'Admin access covers technician rosters, room-level queue analytics and the AI prediction engine for the cardiology ECHO unit.' At the bottom, three statistics are displayed: '6 Echo rooms', '24 Technicians', and '92% AI accuracy'.	 The right panel contains the sign-in form. It starts with a purple 'ADMIN ACCESS' tab. The heading is 'Sign in to ECHO Admin console' with a subtext 'Restricted to authorised hospital administrators.' The form includes three input fields: 'ADMIN ID' (containing 'ECHO-ADM-014'), 'PASSWORD' (masked with dots), and 'TWO-FACTOR CODE' (containing '6-digit authenticator code'). A large teal 'Sign in →' button is at the bottom. Below the button is a link: 'Not an admin? Go to staff sign in'.

Split-panel sign-in screen for the ECHO Unit Admin console.

Purpose

This is the dedicated authentication gate for the Administrator Web Portal. It restricts access to authorised hospital administrators before any dashboard, staff, or room data can be viewed, addressing the security and data-privacy requirements discussed in the Methodology chapter (Section 3.10) of the proposal.

Key Functions & Interactions

- Left panel: brand/marketing copy — "Oversee every echo room, technician and AI-predicted queue in one place" — communicating the scope of the admin console.
- Admin ID field for entering the administrator's staff identifier (e.g., ECHO-ADM-014).
- Password field (masked).
- Two-factor access code field (a 6-digit authenticator code), implementing the two-factor authentication control referenced in the Settings screen and in Section 3.10 (ethical/security considerations).
- "Sign in" call-to-action button that authenticates the administrator and routes to the Dashboard (Screen 2).

Correspondence to the Research Proposal

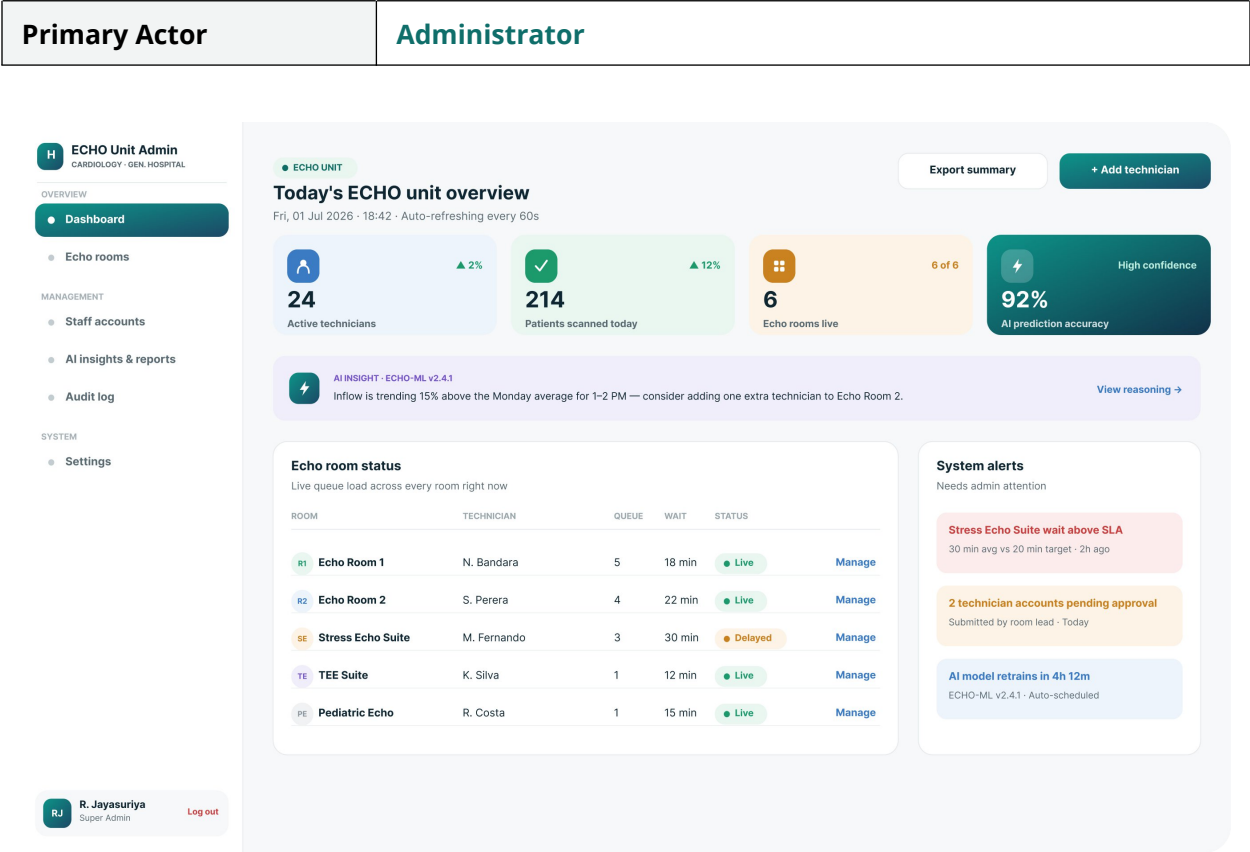
The proposal does not list a login screen among the eight named admin screens, but it is a necessary precondition for all of them and directly implements the security/2FA requirement discussed in Section 3.10 and configured on the Settings screen.

Notes / Corrections

Observed discrepancy

This screen is admin-only (it explicitly says "Sign in to ECHO Admin" and requires an Admin ID). It should not be confused with Screen 12, a separate, shared "Welcome back" login screen with Patient/Staff/Admin tabs found later in the same mockup strip. Two different sign-in designs exist in the file — only one (this one) is exclusively for administrators.

Screen 2: Today's Unit Overview (Dashboard)

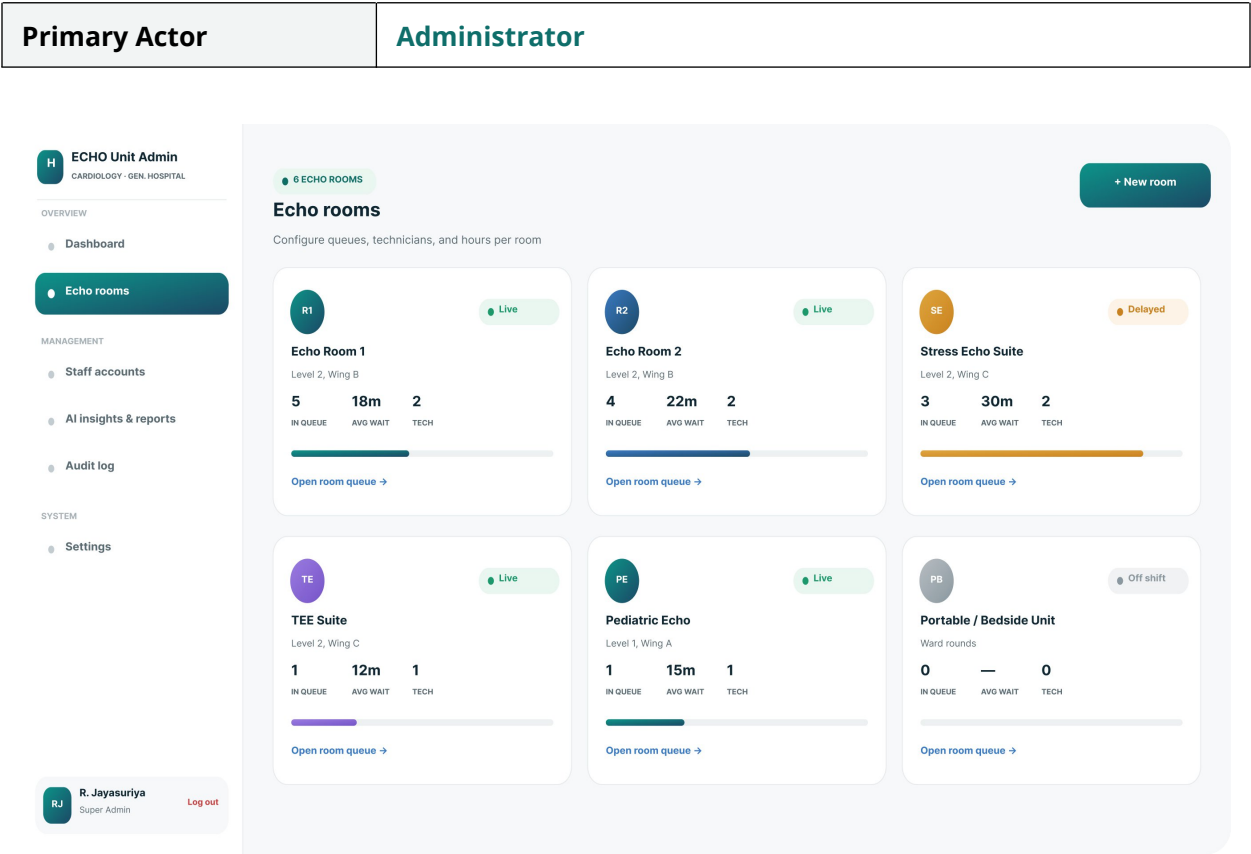


- A live queue snippet (queue number, wait time, status such as Live/Delayed, and a 'Manage' link into that room's queue).
- System Alerts panel flagging issues needing admin attention, e.g. an SLA breach on the Stress Echo Suite, or technician accounts pending approval.

Correspondence to the Research Proposal

This is the 'Today's Unit Overview dashboard combining unit-wide KPI cards, station status, and system alerts' described in Section 1.5 (Scope of the Study) as the first of the eight Administrator Web Portal screens.

Screen 3: Echo Rooms (Room/Station Management Grid)

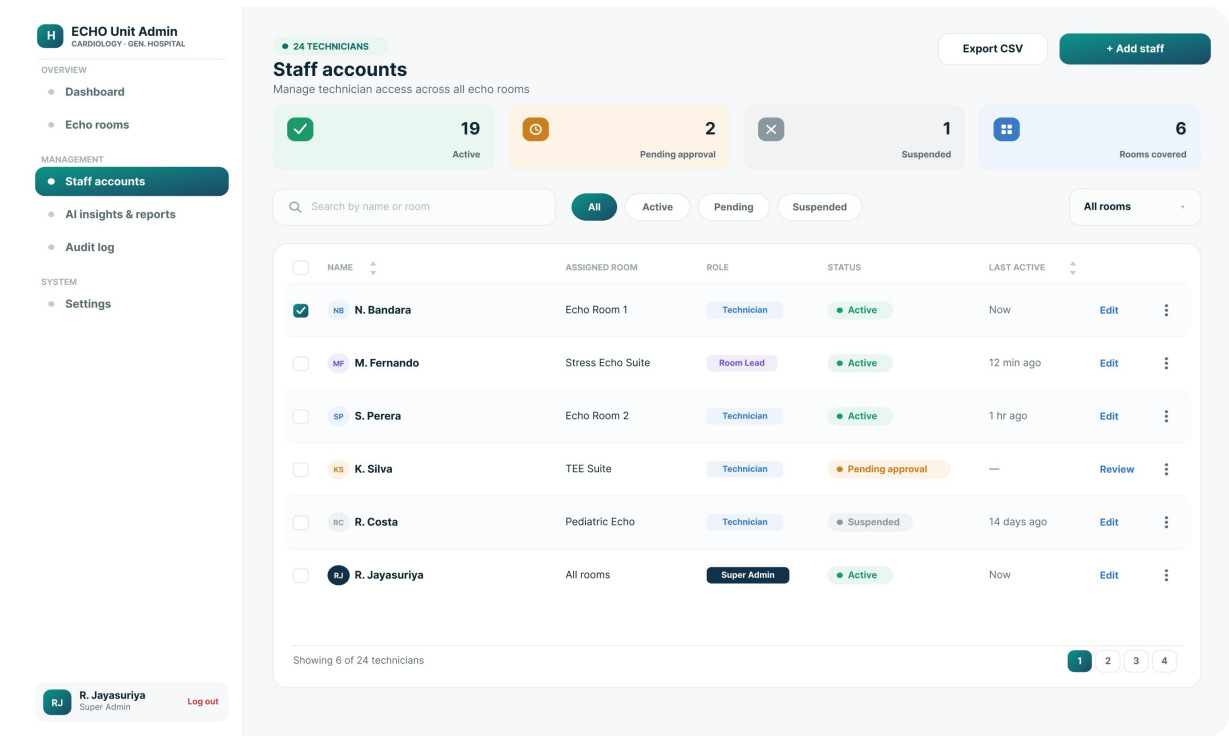


Correspondence to the Research Proposal

This is the 'Room/Station Management grid,' the second of the eight Administrator Web Portal screens listed in Section 1.5.

Screen 4: Staff Accounts

Primary Actor	Administrator
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Searchable, filterable table of ECO unit staff accounts.

Purpose

Centralises staff-account governance for the ECO unit, replacing the informal, undocumented tracking of who is rostered where — addressing the staff-shortage/administrative-overload problem from a governance angle (Chapter 5, Section 5.5).

Key Functions & Interactions

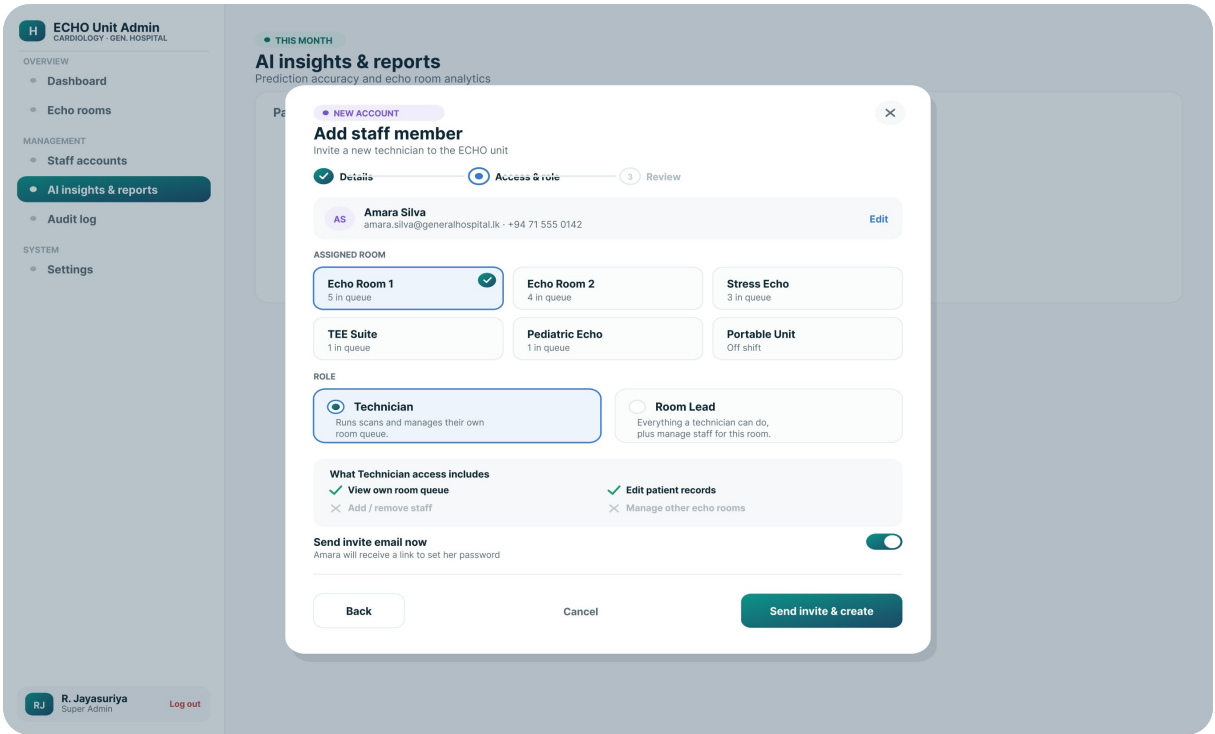
- Search bar to find staff 'by name or room'.
- Filter chips: All / Active / Pending / Suspended.
- Selectable rows (checkboxes) for bulk actions, plus per-row 'Edit' and an overflow (kebab) menu for status actions.
- Columns for staff name, role tag (e.g., Technician, Room Lead), assigned room, account status, and last-active timestamp.
- A pending-approval row surfaces new technician sign-ups (linked to the 'accounts pending approval' alert on the Dashboard) with a 'Review' action.
- Entry point into the Add Staff Member modal (Screen 10) and the staff Access & Role modal (Screen 5).

Correspondence to the Research Proposal

This is the 'Staff Accounts management table', the third of the eight Administrator Web Portal screens listed in Section 1.5.

Screen 5: Staff Member — Access & Role Modal

Primary Actor	Administrator
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Three-step (Details / Access & role / Review) modal for editing a specific staff member's room assignment and permissions.

Purpose

A detailed, tabbed dialog opened from a row on the Staff Accounts screen (Screen 4). It lets an administrator review or change an individual staff member's room assignment and permission level, operationalising the role-based-access-control requirement discussed in Section 3.10.

Key Functions & Interactions

- Step indicator across three stages: Details, Access & role, Review.
- Staff identity strip showing name, email, and phone, with an 'Edit' shortcut.
- Assigned Room selector — a 6-tile grid (Echo Room 1/2, Stress Echo, TEE Suite, Pediatric Echo, Portable Unit) with the current room highlighted and its live queue count shown for context.
- Role selector — Technician ('Runs and manages their own room queue') vs Room Lead ('Everything a technician can do, plus manage staff for this room').

- A live-updating 'What [role] access includes' checklist (e.g. view own room queue, edit patient records vs. add/remove staff, manage other echo rooms) so the admin can see the practical effect of the role before saving.

Correspondence to the Research Proposal

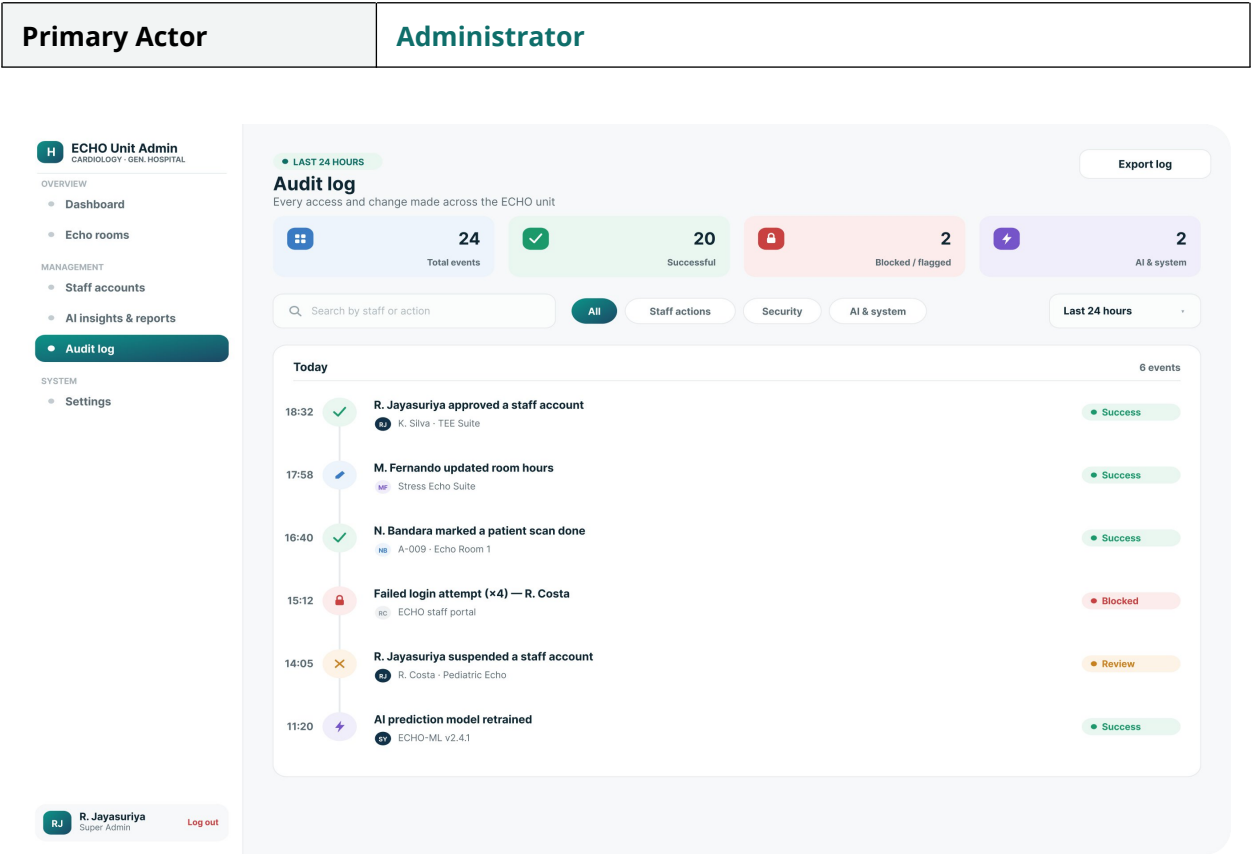
This screen extends the 'Add-Staff-Member modal' named in Section 1.5 into a fuller staff-management dialog; the underlying role/permission model is the same one exposed system-wide on the Settings screen (Screen 7).

Notes / Corrections

Observed discrepancy

The proposal's scope section names only one staff modal ('Add-Staff-Member modal'). The mockups actually show two different staff dialogs: this detailed, tabbed Access & Role editor (used to review/edit an existing staff member such as Amara Silva) and a separate, shorter creation form (Screen 10). They should be read as two states of the same staff-management flow — edit vs. create — rather than two unrelated screens.

Screen 6: Audit Log



Timestamped, user-attributed activity timeline with category and time-range filters.

Purpose

Provides an accountable, tamper-evident record of who did what and when across the system — the data-governance control explicitly called for in Section 3.10 (Ethical Considerations) to support accountability in a system that handles patient data under Sri Lanka's Personal Data Protection Act, No. 9 of 2022.

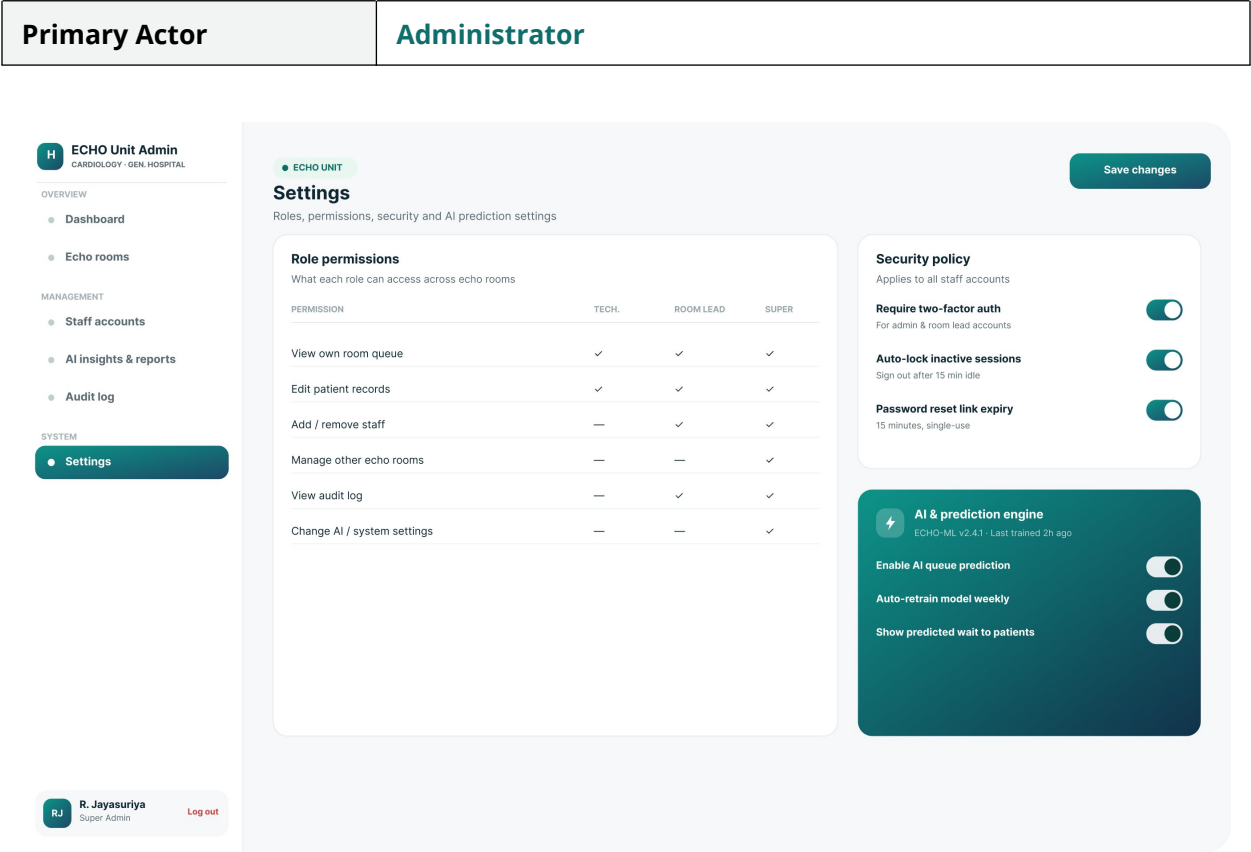
Key Functions & Interactions

- Search bar to find a specific staff member or action.
- Category filter chips: All, Staff actions, Security, AI & system.
- A time-range selector (e.g. 'Last 24 hours').
- A chronological, timestamped timeline grouped by day ('Today'), each entry showing the actor, the action taken (e.g. 'approved a staff account', 'updated room hours', 'marked a patient scan done'), and the affected room/record.
- Status-coloured icons distinguishing successful actions from security events such as a failed login attempt.

Correspondence to the Research Proposal

This is the 'Audit Log,' the fifth of the eight Administrator Web Portal screens listed in Section 1.5, and it is also referenced directly in Section 3.10 as recording 'a timestamped, user-attributed history of system actions for accountability.'

Screen 7: Settings



Role-permission matrix, account-security controls, and AI/prediction-engine configuration.

Purpose

The system-configuration hub for role-based access control, session/security policy, and the machine-learning prediction engine — again tying directly back to Section 3.10's requirement that access to patient data be restricted on a need-to-know basis according to staff role.

Key Functions & Interactions

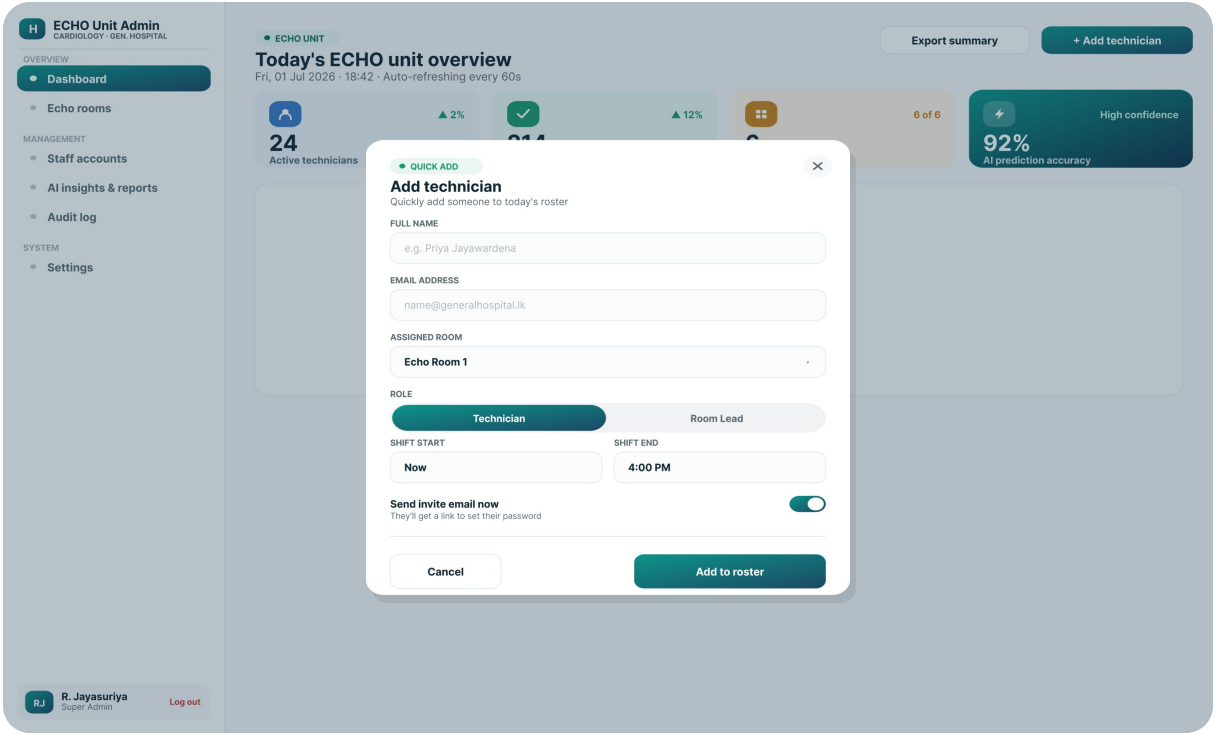
- A permissions matrix cross-referencing capabilities (view own room queue, edit patient records, add/remove staff, manage other echo rooms, view audit log, change AI/system settings) against the three roles (Technician, Room Lead, Super Admin).
- Security controls: Require two-factor authentication (toggle), Auto-lock inactive sessions (with an idle timeout, e.g. 15 minutes), and Password-reset-link expiry (e.g. 15 minutes, single-use).
- An 'AI & prediction engine' panel showing the active ECHO-ML model version and last-retrained time, with toggles for Enable AI queue prediction, Auto-retrain model weekly, and Show predicted wait to patients.

Correspondence to the Research Proposal

This is the 'Settings screen for role-based permissions and security configuration,' the sixth of the eight Administrator Web Portal screens listed in Section 1.5, and its AI-engine controls correspond to the Machine Learning Prediction Service described in Section 3.5 (System Architecture).

Screen 8: Add Technician (Quick-Add Modal)

Primary Actor	Administrator
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Lightweight modal for quickly adding a technician to today's roster.

Purpose

A fast-path dialog for adding a technician to the day's roster on the spot (for example, to cover an unplanned absence flagged in a Dashboard alert), rather than the fuller onboarding flow in Screen 10.

Key Functions & Interactions

- Full Name and Email Address fields.
- Assigned Room dropdown (defaults to a specific room, e.g. Echo Room 1).
- Role toggle: Technician / Room Lead.
- Shift Start and Shift End time fields (e.g. Now → 4:00 PM), scoping the addition to a single shift.
- 'Send invite email now' toggle so the technician receives credentials immediately.

Correspondence to the Research Proposal

This is a same-day variant of the 'Add-Staff-Member modal' named in Section 1.5, oriented around shift coverage rather than permanent onboarding.

Notes / Corrections

Observed discrepancy

This is a distinct, simpler dialog from the official Add Staff Member modal in Screen 10 (it adds Shift Start/End fields and drops the Staff ID field). The two should not be treated as the same screen — they represent two different creation flows (quick same-day add vs. formal account creation).

Screen 9: Add Echo Room Modal

Primary Actor	Administrator
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The screenshot shows the 'Add echo room' modal in the ECHO Unit Admin interface. The modal is titled 'Add echo room' and includes the following fields and options:

- ROOM NAME:** Text field with placeholder 'e.g. Echo Room 3'.
- ROOM TYPE:** Radio buttons for 'Standard Echo' (selected), 'Stress Echo', 'TEE', 'Pediatric', and 'Portable'.
- LOCATION / WING:** Text field with placeholder 'e.g. Level 2, Wing B'.
- OPERATING HOURS:** Two text fields for start and end times, with '8:00 AM' and '4:00 PM' entered.
- LEAD TECHNICIAN:** Dropdown menu with 'Assign later' selected.
- Make room live immediately:** Toggle switch (checked).
- PREVIEW:** A panel on the right showing a preview of the room card. It displays 'R3', 'Echo Room 3', 'Level 2, Wing B', '0' in the QUEUE column, '1' in the TECH column, and a 'Live' status indicator.

At the bottom of the modal are 'Cancel' and 'Create room' buttons. The background interface shows a sidebar with 'ECHO rooms' selected and a '+ New room' button in the top right.

Form for registering a new echo room/station, with a live preview of the resulting room card.

Purpose

Lets an administrator register a new physical room or station and configure its operating parameters, directly supporting the resource-planning objective in Section 1.3.2 ('design and implement administrator-level oversight tools, including station/room management...').

Key Functions & Interactions

- Room Name text field (e.g. 'Echo Room 3').
- Room Type selector: Standard Echo, Stress Echo, TEE, Pediatric, or Portable.
- Location/Wing text field (e.g. 'Level 2, Wing B').
- Operating Hours start/end time fields (e.g. 8:00 AM – 4:00 PM).
- Lead Technician dropdown to assign an initial technician.
- 'Make room live immediately' toggle.
- A live Preview panel on the right showing exactly how the new room card will appear on the Echo Rooms grid (Screen 3) before it is saved.

Correspondence to the Research Proposal

This is the 'Add Echo Room modal,' the eighth of the eight Administrator Web Portal screens listed in Section 1.5.

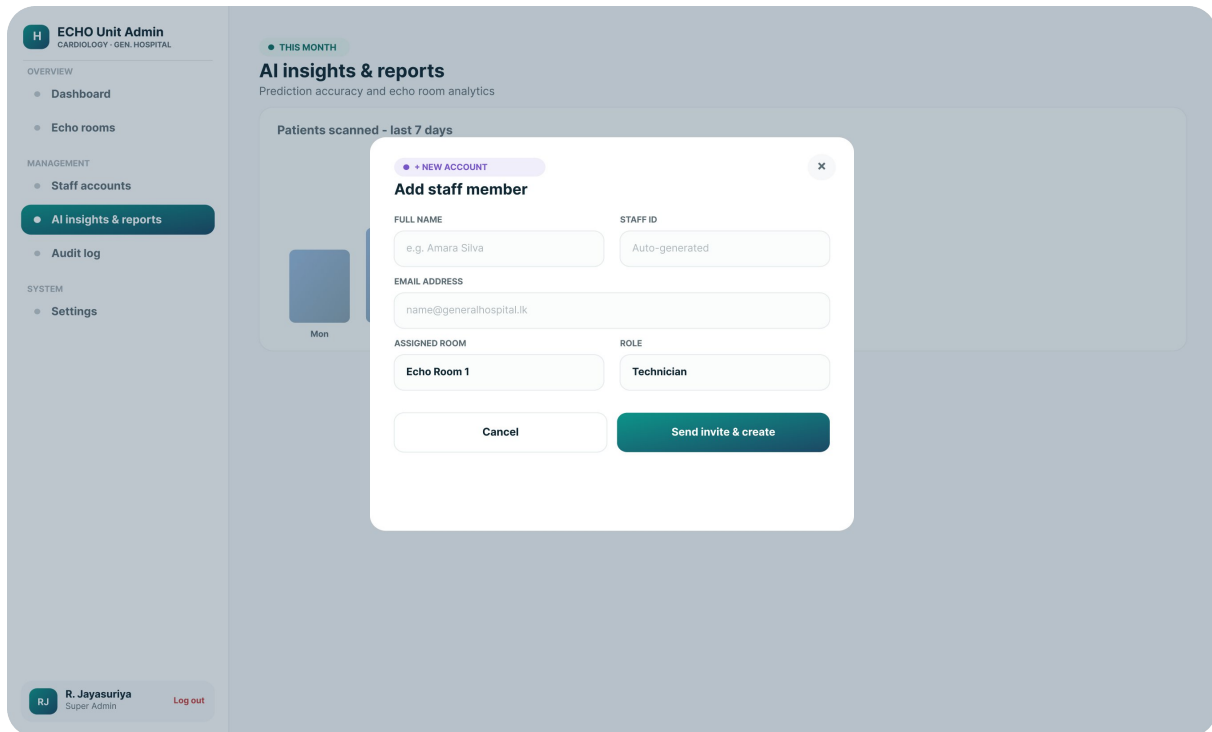
Notes / Corrections

Observed discrepancy

Section 1.5 lists a 'Register Station modal' and an 'Add Echo Room modal' as two separate screens. Only one room/station-creation modal appears in the mockups, and its Room Type options (Standard Echo, Stress Echo, TEE, Pediatric, Portable) cover both diagnostic rooms and general stations in a single form. In the documentation and any future write-up, this should be described as one combined 'Add Echo Room / Register Station' modal rather than two separate implemented screens, since no second, distinct 'Register Station' form exists in the provided mockups.

Screen 10: Add Staff Member Modal

Primary Actor	Administrator
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Formal staff-onboarding modal, opened from the Staff Accounts screen.

Purpose

The formal path for onboarding a new permanent staff account into the ECO unit, as opposed to the same-day 'Add Technician' shortcut in Screen 8.

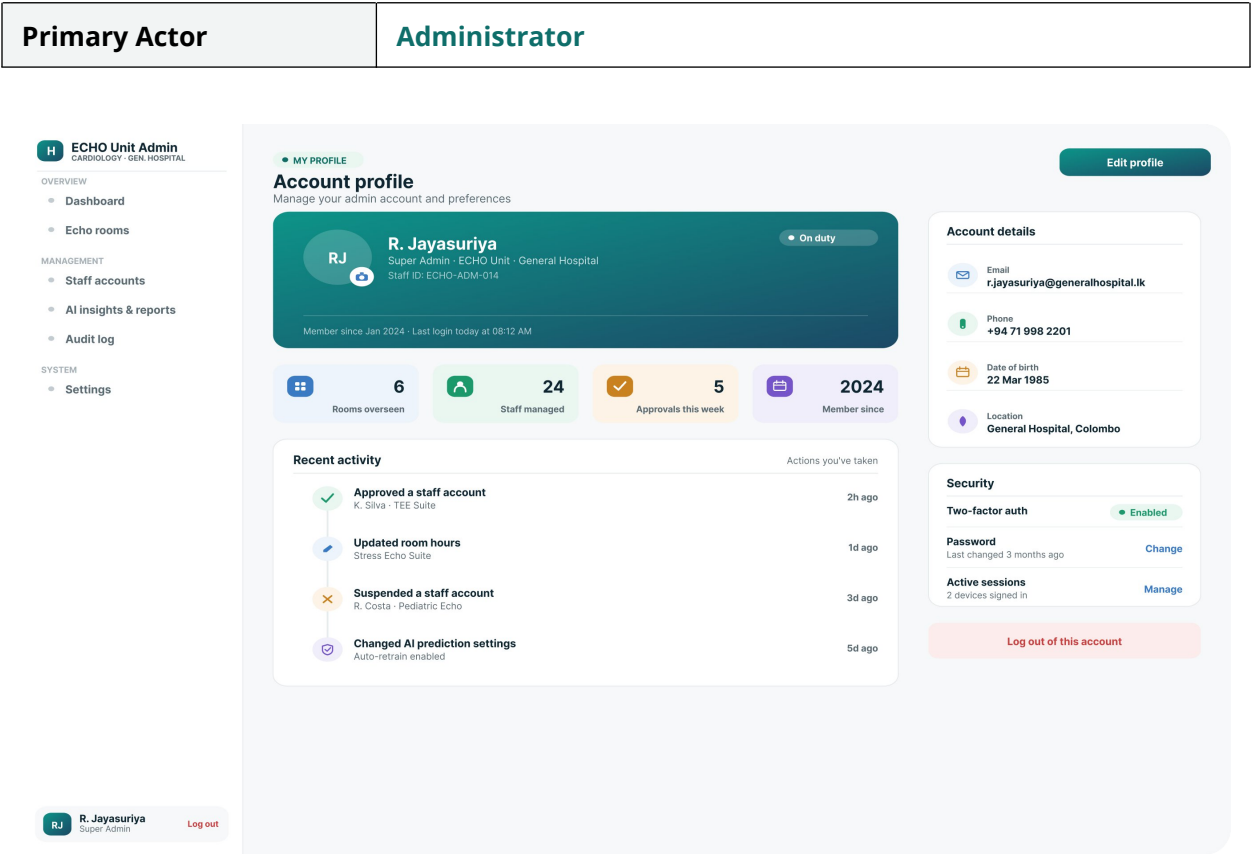
Key Functions & Interactions

- Full Name and auto-generated Staff ID fields.
- Email Address field (used for the invitation).
- Assigned Room and Role dropdowns.
- Cancel button to dismiss without saving.
- 'Send invite & create' primary action, which creates the account and emails the new staff member their sign-in credentials.

Correspondence to the Research Proposal

This is the formal implementation of the 'Add-Staff-Member modal,' the fourth of the eight Administrator Web Portal screens listed in Section 1.5.

Screen 11a: My Profile (Admin Account Overview)



The logged-in administrator's own account summary, statistics, and recent activity.

Purpose

Gives the Super Admin a personal account summary and quick self-service access to their own security settings, separate from the unit-wide Dashboard (Screen 2).

Key Functions & Interactions

- Profile banner showing membership date and last login time.
- Personal stat cards: rooms overseen, staff managed, approvals this week, and member-since year.
- A 'Recent activity' feed of the admin's own actions (e.g. approved a staff account, updated room hours, suspended a staff account, changed AI prediction settings).
- A contact/details card (phone, date of birth, hospital location).
- A Security card exposing two-factor-auth status, a 'Change' password link, 'Manage' active sessions, and a 'Log out of this account' action.

Correspondence to the Research Proposal

Not individually named in Section 1.5's list of eight admin screens, but it is a normal supporting screen for any authenticated web portal and reuses the same security controls (2FA, sessions) described in Section 3.10.

Notes / Corrections

Observed discrepancy

This screen is not one of the eight screens explicitly scoped in the proposal. It should be documented as an additional, supporting Administrator screen rather than mapped onto any of the eight named ones.

Screen 11b: Edit Profile

Primary Actor	Administrator
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ECHO Unit Admin

CARDIOLOGY - GEN. HOSPITAL

OVERVIEW

- Dashboard
- Echo rooms

MANAGEMENT

- Staff accounts
- AI insights & reports
- Audit log

SYSTEM

- Settings

RJ

R. Jayasuriya

Super Admin

Log out

MY PROFILE

Edit profile

Update your personal details and work information

Profile photo

JPG or PNG, at least 200x200px

RJ

Remove

Upload new

Personal information

FULL NAME

R. Jayasuriya

PHONE

+94 71 998 2201

EMAIL ADDRESS

r.jayasuriya@generalhospital.lk

DATE OF BIRTH

22 Mar 1985

Work details

STAFF ID

ECHO-ADM-014

LOCKED

ROLE

Super Admin

LOCKED

DEPARTMENT

Cardiology - ECHO Unit

LOCKED

ACCESS LEVEL

All echo rooms

LOCKED

Managed by system - not editable

Cancel

Save changes

Editable personal-information and work-details form, reached from My Profile.

Purpose

Lets the administrator update their own personal contact details, while keeping system-managed fields (role, access level, department) locked for integrity.

Key Functions & Interactions

- Editable Personal Information: full name, email address, phone, date of birth.
- Work Details section: Staff ID and Department fields (both locked, labelled 'Managed by system — not editable'), plus Role (e.g. Super Admin) and Access Level (e.g. All echo rooms), also locked.
- Save/Cancel actions in the header (implied by the button pair shown above the form).

Correspondence to the Research Proposal

Companion screen to My Profile (11a); not separately named in Section 1.5.

Notes / Corrections

Observed discrepancy

As with 11a, this is an additional screen beyond the eight explicitly scoped in the proposal and should be documented as a supporting account-management screen rather than forced into one of the eight named categories.

Screen 12: Unified Sign-In (Role Selector)

Primary Actor	Shared — Patient, Staff, and Administrator
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System-wide 'Welcome back' sign-in screen with Patient / Staff / Admin role tabs.

Purpose

This is the single, shared entry point for the entire ECHO system described in Section 3.5 (System Architecture): 'three separate client applications are provided for the three user roles ... all three clients communicate with a shared backend server.' Rather than being an admin-only screen, it is the front door that routes a person into whichever of the three portals matches their selected role.

Key Functions & Interactions

- A left-hand hero panel ('One secure door for every role on the way...') explaining that patients, sonographers, cardiologists, and unit admins all sign in through the same verified gateway, with access scoped the moment they log in, plus trust indicators (role types supported, 256-bit encryption, 24/7 uptime).
- A role selector with three tabs: Patient, Staff, and Admin — the screenshot captures the Patient tab active by default.
- NIC Number and Phone Number fields used for patient sign-in/verification (fields would differ for the Staff and Admin tabs, e.g. staff ID / admin ID and password, as seen in Screen 1).

- 'Remember this device' checkbox and a 'Trouble signing in?' recovery link.
- A 'Continue' call-to-action, and a prompt for new patients to register with their NIC in one step.

Correspondence to the Research Proposal

This screen is not part of the Administrator Web Portal's eight scoped screens; it corresponds instead to the shared authentication layer of the overall system architecture in Section 3.5, and to the patient-registration process referenced for the Patient Mobile Application in Section 1.5.

Notes / Corrections

Observed discrepancy

This is the single most important correction to the original 'all screens are admin' assumption: this screen is explicitly multi-role (it has Patient / Staff / Admin tabs and defaults to Patient), so it must not be classified as an Administrator screen. It should be documented separately as the shared system-login/gateway screen, distinct from the dedicated Admin Sign-In screen (Screen 1).

3. Summary of Corrections Made to the Original Framing

- Not all twelve screens in the mockup strip are Administrator screens. Screen 12 ('Welcome back', with Patient/Staff/Admin tabs) is a shared, role-selecting login gateway used by all three portals, not an admin-only screen.
- There are two different sign-in designs in the file: a dedicated Admin Sign-In (Screen 1, admin ID + password + 2FA code) and the shared, tabbed role-selector login (Screen 12). These should be documented as two separate screens, not duplicates of each other.
- The proposal's Section 1.5 names a 'Register Station modal' and an 'Add Echo Room modal' as two separate screens, but the mockups only contain one room/station-creation modal (Screen 9), whose Room Type field already covers standard, stress, TEE, pediatric, and portable configurations. This should be reported as a single combined modal, not two distinct implemented screens.
- The single 'Add-Staff-Member modal' named in the proposal is, in the mockups, actually three related but distinct dialogs: a formal creation form (Screen 10), a same-day quick-add shortcut (Screen 8), and a fuller Access & Role editor for existing staff (Screen 5). All three should be documented individually rather than collapsed into one screen.
- Two supporting account screens — My Profile (11a) and Edit Profile (11b) — exist in the mockups but are not named among the eight core Administrator Web Portal screens in Section 1.5. They should be reported as additional screens rather than omitted or mis-mapped onto one of the eight.